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An Internet of Things based Waste Management System using Hybrid Machine Learning Technique

1Arunkumar M S
arunkumarmsster@gmail.com
School of Computing
Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology

[2]Sathishkumar P
mailtostp@gmail.com
Department of Computer Scienec and Engineering
Bannari Amman Institute of Technology

[3]Suguna R
sugunarajasekar@gmail.com
Department of Computer Scienec and Engineering
Bannari Amman Institute of Technology

[4]Deepa S
sdeepakec@gmail.com
Department of Computer Technology
Kongu Engineering College

***Abstract*— The most significant aspects of creating smart cities is waste management. Recycling and landfilling are two methods of waste management that lead to the demolition of trash. Because of population expansion, it is difficult to maintain cleanliness in urban areas. Because the machine learning (ML) and Internet of Things (IoT) eases the gathering, integration, and processing of diverse kinds of information, it provides an agile solution for classification and real-time monitoring. It is our intention to create a waste management scheme based on the IoT. The IoT has been used to keep tabs on people's movements and to help with garbage management. A machine learning technique called Decision Tree with Extreme Learning Machine was used to analyze data about a city (DT-ELM). The single classifier requires iterative training, which is time consuming, but the suggested hybrid model does not. Decision trees use traits that are good at classifying. Additional weights for the selected features are calculated to improve their categorization accuracy. We use the entropy theory to map the decision tree to ELM in order to get accurate prediction results. The garbage kind, truck size, and waste source may all be analyzed thanks to the network. In order to take the proper action, the waste management centers were informed of this information. An experiment was conducted to test the efficiency of an IoT-based trash management system.**

Keywords*— *Waste Management; Machine Learning; Internet of things; Data Monitoring; Recycling of Waste

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I. Introduction

From the time of its genesis to the time of its demolition, waste management encompasses all of the necessary operations and actions. A waste product can be in any of these three states: liquid, gas, or solid. Many different methods are used to deal with various kinds of garbage: biological, industrial, and even residential waste. Cardboards, plastics, papers, glasses, biowaste, and other household items are examples of household waste. There is no way to recycle or classify domestic trash into biological or material categories [1-2]. There is an urgent need to identify trash and apply suitable treatments (e.g. composting, incineration, landfilling for various kinds of waste in light of the rapid development in resource use and the increasingly serious environmental challenges. Waste should be separated as soon as possible to maximise the quantity that may be recycled and minimise the risk of contamination [3].

The IoT is a promising novel technology that, because of its wide range of interconnectivity, has the possible to positively alter human life on a worldwide scale. Connecting low-energy devices and people via the Internet is facilitated by the IoT. New services for smart cities [4] and improved energy efficiency have been implemented by several applications around the world using IoT-based operations. Smart city growth based on the IoT is having an impact on urbanism [5]. Smart cities improve people's quality of life and their relationships with the natural environment. Intelligent smart cities use information and communication technology (ICT) to decrease urban waste and improve operational efficiency. Infrastructure (concurrent waste reduction and energy management consumption [6] are all part of the smart city components. [7] The inclusion of IoT devices enhances this type of system. While developing smart cities, IoT devices are linked to the internet [8].

One of the most important aspects of artificial intelligence (AI) is machine learning (ML), which is the process through which computers may automatically learn and make decisions on their own [9]. Models and algorithms in machine learning are studied in a scientific manner. ML's popularity has

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skyrocketed as a result of its unique set of capabilities. Regression, classification, clustering, and correlation rule perception [10] are some of the useful ML methods for IoT-based trash management. For the following three main causes: IoT apps acquire a huge amount of data every day because all devices are connected. It's possible that they can be trained to perform certain actions based on specified conditions or interesting feedback from the data they've gathered so far. Next is the ability of software programmes, like as classifiers and clusters and forecasters to learn how to execute specific tasks. Many algorithms and statistical models can be used to train these systems to analyse sample data. It's also common for ML algorithms to try to uncover connections between measurable properties (called features) and some output values (called labels). After training, you can use the information you've learned to look for trends or make decisions based on fresh facts. In most cases, human interaction is required to analyse the calm data, extract important information, and develop smart applications. Devices in the Internet of Things (IoT) must be self-sufficient in addition to collecting and transmitting data. Context-based decision-making and data-gathering are essential. Waste management schemes utilising IoT-based machine should therefore take into account additional variables when predicting tasks.

Hybrid machine learning techniques (DT-ELM) are utilised to classify various wastes, which will be discussed in more detail later on. Although varied topologies lead to different predictions. We use the entropy theory to translate the DT construction to the ELM structure in order to get accurate waste management outcomes. In ELM, the mapping rule specifies not only the amount of neurons in each layer, but also the connections among the input and hidden layer. As a result, ELM can begin with an appropriate assignment of information. A number of different performance indicators are tested on TrashNet datasets by altering the data. The training time of proposed model is average of 4s, because all important features are already extracted. The rest of the document is structured as follows: Section 2 focuses on the current literature, while Section 3 delivers a brief discussion of the proposed methodology. Section 4 contains the results of the experiments, and Section 5 depicts the study's conclusion..

II. Related Works

The electrical and electronic equipment waste from photographs was explored by Nowakowski and Pamua [11] using an image recognition system. E-waste collection businesses establish a collection strategy by assigning enough vehicles and payload capacity for a certain e-waste project based on the quantity and kind of waste. Other researchers have attempted to present sensors to increase classification performance in addition to image classification.

Experimenting with various alternative deep CNN architectures and different optimization methods, Bircanoglu et al. [12] selected the TrashNet dataset. Improved models based on existing models aim to reduce forecast time by modifying the connection designs of skip influences in packed blocks. Because of the use of transfer learning, the DenseNet121 model achieved the best results with 95% accuracy. Additionally, in order to accurately detect the presence of glass and metal in garbage bags, Funch and colleagues [13] employed sound recordings and a beat-frequency oscillation metal detector, which they fed into CNNs. They reported a detection rate of 98 percent.

Additionally, the MWM system has benefited from data exchanges and new analytical methodologies in addition to improved waste sorting procedures. According to Niska and Serkkola [14], modern data analytic methodologies can produce more specific waste generation information for planning and optimising garbage collection and recycling, based on the findings of their experimentation. An IoT-Based Urban Waste Management System developed by Alqahtani et al. [15] uses a cuckoo search-optimized LSTM to monitor human activity and collect and process urban information for the waste management centre to analyse so that appropriate actions can be taken.

The IoT-based solid waste management system created by Idwan and colleagues in [16] used routing algorithms to find the best paths for the garbage truck fleet. Reduced traffic and quick removal of garbage were the primary objectives. The algorithm used to determine the best path used a variety of options. The most effective garbage collection route was selected with the help of a genetic algorithm operator. An image-based trash management system was developed by Ruiz et al. in [17] using CNN. Analyses were conducted using images from the TrashNet database. Image processing was done using convolutional networks such the RESNET family, as well as inception and VGG. Up to 88.5 percent of the time, these networks were able to correctly classify garbage by using picture attributes they learned from training data.

III. Proposed Methodology

Waste classification by machine learning (ML) and the building of smart garbage bins that can collect real-time data via IoT are the two components of the suggested methodology. Waste management can be improved by combining two structural models. The accurate categorization of garbage aids in the identification of reusable waste. We can use recyclable waste without it degrading if we can identify it. Reducing the usage of recyclable components has prompted the authors to apply ML for trash categorization while monitoring waste to distinguish recyclable wastes.

A. Dataset Description

Using photos from the TrashNet database [17], we were able to categorise the incoming waste items. Plastic, glass, cardboard, paper, waste, and metal were all included in the data collection. There were 2,527 images in total, including 501 images of glass, 594 images of paper, 482 images of plastic, 403 images of cardboard, 137 images of rubbish, and 410 pictures of metal. Sunlight and poster board were used for room lighting to shoot these trash photographs. In order to identify waste categories, the dataset was pre-processed, and the scaled photos were used. FIGURE 1: EXAMPLES OF TrashNet IMAGES.

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Figure 1: Sample images

Additional IoT devices are required to collect data from TrashNet in order to recover urban waste organisation. Smart towns employ smart bins to collect waste. The control room receives information in overflow or empty states. Waste collection trucks can be deployed based on this information. The smart city's garbage is collected and processed using a proposed model. A fixed camera captures the discarded photographs. Because image has a sole signature or property, the features are retrieved from the collected images. Images can be more easily distinguished from one another when their derived attributes are used.

B. *Feature Extraction*

In this proposed system consist of three different kind of feature extraction such as Color, Shape and texture features.

1) *Extraction of Color Features*

In order to improve performance, a technique known as feature fusion or feature combining is performed. That's because just a small number of features (e.g., translation, rotation, and lighting changes) are able to withstand the considerable changes that occur in the database of images. This paper proposes a method based on colour feature extraction. In terms of features, colour is one of the most extensive, and it's linked to the image's background and foreground objects, respectively. As a result, colour is resistant to rotation and translation. There are other applications for colour characteristics as well. Color models and mapping functions are both recognised in the colour space.

After detecting the colour edge, the initial and final photos are shown. In this case, the distance between the feature vector of the image supplied as a query and the database images is taken into consideration. A comparable image will be selected in this case, if the F as query image vector feature's distance from database images is small enough to be considered a similar image. This type of search relies on parity rather than exact match. It is also necessary to use the index of analogy in order to calculate the Manhattan distance. As a result, the results of the search are ranked using the same index.

$$D = \sum_{I=1}^{n} |F_q - F_r| \qquad (1)$$

2) *Extraction of Shape Features*

An image's shape qualities (such moment, region, and border, for example) are captured by extracting shape features. Shape can be stored, transmitted, compared, and identified with this method. Feature robustness to rotation, translation, and scaling is critical. In order to store, transmit, or recognise shape, researchers have investigated the best methods for discovering the features of shape. In this sense, the shape features chosen do not undergo any mathematical modification. The RGB colour image contains three values for each pixel, while the grayscale image is used to extract the form features. The formula projected by Craig is employed as stated below for this purpose:

$$I_g = [I_r I_g I_b] * [0.29890.114] \qquad (2)$$

It follows that I g is the joint 2D matrix; I r is the 2D matrix. Components of colour are represented by I b, and grey-level images are represented by I g. The median filter is used in the pre-processing step to reduce noise. It is possible to reduce speckling, pepper and salt noise by using a median filter. Because it can preserve fine details and crisp edges in digital photos, the median filter is often used to avoid softening the edges. The median filter removes nearly all of the image's noisy data. Since pixels with very similar values are separated using the neutrosophic clustering approach and indeterminate pixels are ignored when processing the grey image,

The canny edge method is utilised to complete the assignment. This is useful for determining the boundaries of adjacent pixels that are otherwise identical. The shape features were extracted using the canny edge approach in this study. Various shapes in the I g image can be extracted using the clever edge detection algorithm. The next step is to extract the shaped content indices. Features vector indices are kept in the database.

3) *Extraction of Texture Features*

Gray-Level Co-Occurrence Matrix (GLCM), looks to be a reliable and successful method for classifying images using statistical analysis, and it can be represented as a matrix consisting of two dimensions of combined similarity between pairs of pixels, with a distance d between them in a defined direction. These findings appear to be substantially connected in some way. The decision to use fewer features was based on the notion that including characteristics that aren't important reduces model accuracy and raises training costs. As a result, selecting a large number of features can upsurge the training time for models.

C. *Waste material identification using proposed DT-ELM Algorithm*

Decision Layer uses a decision tree with a maximum information gain ratio to execute feature extraction. D is the decision layer's input. Class lable x i is a list of x i[x i1,x i2,...x ir]Rr for each instance in D, where x i is the class of x i. The decision layer's output, D', only contains the characteristics that were explicitly chosen for inclusion. If you look at the x inew, you can see that each of the instances in X is a list of n features from D'.

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Determination trees are built by partitioning the data set into smaller subsets, for example by making sure that all the subsets fit to a single class. For each node in the tree, a single feature split is defined recursively using some criterion. The info gain ratio is an extensively used decision tree criterion. As a result of information theory, entropy is defined as

$$Info(D) = -\sum_{c=1}^{m} p_c log_{\,2}(p_c) \quad (3)$$

The number of classes is represented by m. Is it more likely than not for the class of x i that it belongs to? The predictable decrease in entropy following the split is used to define the split rule, and this decrease is represented by the information gain. The information gain is described as follows.

$$Gain\ (A) = Info(D) - Info_A(D) \quad (4)$$

$Information\ A\ (D)$ indicates the entropy of D when it is partitioned by feature A, which is the conditional entropy of D. The weighted average of all sets arising from the split stated in (5), where |D k |/|D| functions as the k th partition's weight.

$$Info_A(D) = \sum_{k=1}^{K} |D_k|/|D| \times info(D_k) \quad (5)$$

The "split information" value of the information gain ratio extends the information gain and applies a normalisation to the information gain.

$$SplitInfo_A(D) = -\sum_{k=1}^{K} |D_k|/|D| \times log(|D_k|/|D|) \quad (6)$$

In order to choose which feature will be used as the splitting feature, the information gain ratio is calculated as follows.

$$GainRatio(A) = \frac{Gain\ (A)}{SplitInfo_A(D)} \quad (7)$$

This is how the decision tree is built. There is a node in the decision tree for each feature that was selected. The specific class of a categorical variable is represented by a terminal node. Figure 2 presents the block diagram of proposed model.

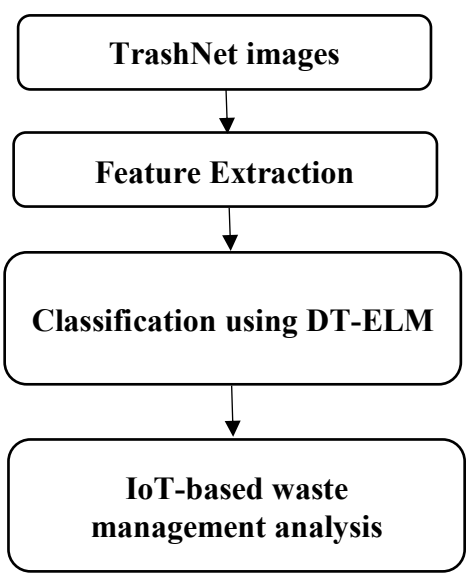

Figure 2: Block diagram of proposed model.

Enhancement Layer. Because each feature's categorization ability varies, the influence on each leaf node will be unique. All leaf nodes are connected to the root node, which has the best classification ability. That means that the root node has the largest impact on all of the leaf nodes in the tree. As a result of there being fewer leaf nodes to connect to, and therefore less influence on, each of the internal nodes, save for the root node. Each picked feature's value is multiplied by the number of leaf nodes it connects to in the decision tree, which is equal to I s. It is depicted in this manner.

$$x_{is_{new}} = x_{is_{ori}} \times I_s, \quad s = 1,2,\ldots,n \quad (8)$$

Mapping Layer. The decision tree is mapped to the ELM in the mapping layer, inspired by the entropy net. Table 1 displays the mapping rules between ELM neurons and nodes in the decision tree.

TABLE 1: MAPPING RULES AMONG DT AND ELM.

DT	ELM
Internal nodes	Input neurons
Paths between nodes	Connections between input layer and hidden layer
Terminal nodes	Hidden neurons
Distinct classes	Output neurons

The sum of neurons in the ELM's input layer is equal to the sum of internal nodes in the decision tree. ELM's hidden layer corresponds to each leaf node in the decision tree. The sum of neurons in the output layer of ELM is equal to the number of distinct classes in the decision tree. The connections between the input layer and the ELM's hidden layer are determined by the decision tree's pathways between nodes. As a result of this approach, the number of neurons in each layer is established. In the meantime, it reduces the number of connections needed to improve ELM.

The decision tree's root node is used to map the input layer's first neuron. It connects to all leaf nodes' hidden neurons. As a result, the first neuron has an effect on all subsequent ones, as well. Using the decision tree structure, the second input neuron connects to the four hidden neurons. As a result, the second neuron has an effect on the four neurons that aren't readily visible. The dashed lines in the decision tree show that there are no connections between the interior nodes and the leaf nodes. When the input and hidden layer of ELM are not connected, there are no connections between neurons in both layers.

D. Internet of things-based waste information analysis

In order to remove any overflowing garbage from an area, waste sources are assessed on a regular basis. Because educational institutions and hospitals account for the majority of the available resources, each type of trash has its own unique identification and ranking. Depending on the urgency, the waste must be collected as soon as possible. Once the smart bin [18] has been connected to the cloud by GPRS or

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GSM, it continuously communicates information about the bin's type and volume of waste and its weight. The smart bin weight is compared to the threshold value each time it is used (90 percent of the weight). The alert message is sent to the control room when the bin reaches the threshold value, so the waste can be collected. The data collected from the smart bin-based waste is then evaluated to help with garbage truck dispatch. Weight, volume, and priority are all taken into consideration when selecting the truck. Additionally, the improved neural network is able to accurately forecast the sorts of garbage that will be generated, making waste collection and disposal easier. With this method, waste can be hauled away quickly, and the city may remain clean at all times. The efficiency of the system will be discussed in the following section.

Techniques	Number of bins	PR	R	F-M	FPR	ACC
ANN	20	97.5	97.03	97.50	11.08	97.90
	40	97.5	97.23	97.85	10.50	97.80
DT	20	97.65	97.56	97.58	10.80	98.20
	40	97.12	97.36	97.01	11.05	97.23
EML	20	97.32	97.54	97.42	11.35	97.55
	40	97.86	97.64	97.35	11.60	97.60
PROPOSED	20	97.90	97.89	97.87	08.30	98.01
	40	97.12	97.64	98.01	08.20	98.05

4. Results and Discussion

A desktop machine with an Intel 2.5GHz CPU and 8GB of RAM is used for experiments in MATLAB R2016b and Python 2.7. LIBSVM and Keras libraries are both utilised in the hybrid ML implementation.

TABLE II: VALIDATION OF PROPOSED MODEL WITH DIFFERENT NUMBER OF BINS

TABLE III: VALIDATION OF PROPOSED MODEL WITH DIFFERENT NUMBER OF BINS

Techniques	No. of bins	PR	R	F-M	FPR	ACC
ANN	20	91.70	91.90	91.95	10.00	92.20
	40	90.25	90.21	90.07	10.05	90.80
DT	20	90.21	90.28	90.75	10.35	90.65
	40	90.45	90.43	90.90	9.60	91.23
EML	20	93.05	93.23	94.07	08.30	94.09
	40	94.24	94.78	94.90	07.20	94.90
PROPOSED	20	94.80	94.85	94.95	06.04	95.05
	40	96.70	96.45	95.67	05.08	97.45

4.1. Performance Metrics

There are numerous solutions to assess FS procedures. The size chosen depends on the nature of the application. In this unit, we define all key performance indicators that are used to appraise the projected method. All selected measurements can be considered using the exit from the confusion table. A confusion table is signified by four main parameters:

- **True Positive (TP)**: Sum of wastes examples classified properly.
- **True Negative (TN)**: Sum of different examples classified properly.
- **False Negative (FN)**: Sum of decomposable waste examples wrongly classified as plastics.
- **False Positive (FP)**: Sum of plastics examples wrongly classified as bio-degradable wastes.

Definition of performance metrics and formulas according to the confusion matrix as follows:

- **Sensitivity**: Measures the percentage of effective wastes from collected bins.
- **Accuracy**: It measures the percentage of correct ranked classes out of the total sum of rankings.
- **False Positive Rate**: It measures the percentage of normal wastes identified as plastics.
- **F-score**: Degree the accuracy of the model by seeing both accuracy and recall.

4.2. Performance Validation of Proposed Model

Here two set of experiments are carried out with proposed method and existing techniques such as varying the training and testing dataset. Table 2 shows the validation of proposed hybrid model with existing techniques by using 80% of training data and 20% of testing dataset.

Table 3 shows the experimental analysis of proposed hybrid model by using 60% of training dataset and 40% of testing dataset.

IV. Conclusion

The goalmouth of this research is to create a society that is environmentally sound and healthy while also being intelligent. An ML-based method to smart trash management was examined in this research. Using the TrashNet dataset, photos of trash were first gathered. After that, the photos were altered. Using comprehensive feature descriptors, we were able to identify certain features of the materials. The characteristics were developed and saved in the database with their relevant class labels based on the derived value. Calculations of the bin weight and volume were effective in

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transmitting data from smart bins to the truck control centre. Using bin priority as an organising principle, the new method helped to enhance waste management as a whole. More sensors, which might be recycled or used, will be employed in hospitals with a variety of specialties, as well as public spaces and businesses to separate various forms of garbage

REFERENCES

[1]. Rautela, R., Arya, S., Vishwakarma, S., Lee, J., Kim, K.H. and Kumar, S., 2021. E-waste management and its effects on the environment and human health. Science of the Total Environment, 773, p.145623.

[2]. Jiang, P., Van Fan, Y. and Klemeš, J.J., 2021. Data analytics of social media publicity to enhance household waste management. Resources, Conservation and Recycling, 164, p.105146.

[3]. Adhithya Prasanna, M., Vikash Kaushal, S., Mahalakshmi, P., 2018. Survey on identification and classification of waste for efficient disposal and recycling. IJET 7, 520.

[4]. Shenoy, Akshata, P. Ranjitha, R. Haripriya, and C. B. Vinutha. "Smart Waste Segregation System using Convolutional Neural Networks." Journal of Electrical Engineering 4, no. 2 (2022): 86-99.

[5]. Hengjinda, P., Joy Iong Zong Chen, and Joy Iong Zong. "Renewable Energy Production from Agricultural Waste and Hydrogen Battery Formation." Journal of

[6]. Electrical Engineering and Automation 2, no. 4: 151-155.Saba, D., Sahli, Y., Berbaoui, B., Maouedj, R.: Towards smart cities: challenges, components, and architectures. In: Hassanien, A., Bhatnagar, R., Khalifa, N., Taha, M. (eds.) Toward Social Internet of Things (SIoT): Enabling Technologies, Architectures and Applications Studies in Computational Intelligence, 846th edn. Springer, Cham (2020)

[7]. Silva, B.N., Khan, M., Han, K.: Towards sustainable smart cities: a review of trends, architectures, components, and open challenges in smart cities. Sustain. Cities Soc. (2018).

[8]. Tanwar, S., Tyagi, S. and Kumar, S., 2018. The role of internet of things and smart grid for the development of a smart city. In Intelligent communication and computational technologies (pp. 23-33). Springer, Singapore.

[9]. Mohanta, B.K., Jena, D., Satapathy, U. and Patnaik, S., 2020. Survey on IoT security: challenges and solution using machine learning, artificial intelligence and blockchain technology. Internet of Things, 11, p.100227.

[10]. M. Cerchecci, F. Luti, A. Mecocci, S. Parrino, G. Peruzzi, and A. Pozzebon, "A low power IoT sensor node architecture for waste management within smart cities context," Sensors, vol. 18, no. 4, p. 1282, 2018.

[11]. Nowakowski, P., Pamuła, T., 2020. Application of deep learning object classifier to improve e-waste collection planning. Waste Manage. 109, 1–9.

[12]. Bircanoglu, C., Atay, M., Beser, F., Genc, O., Kizrak, M.A., 2018. RecycleNet: Intelligent Waste Sorting Using Deep Neural Networks. In: 2018 Innovations in Intelligent Systems and Applications (INISTA). Presented at the 2018 Innovations in Intelligent Systems and Applications (INISTA), IEEE, Thessaloniki, pp. 1–7.

[13]. Funch, O.I., Marhaug, R., Kohtala, S., Steinert, M., 2021. Detecting glass and metal in consumer trash bags during waste collection using convolutional neural networks. Waste Manage. 119, 30–38.

[14]. Niska, H., 2018. Data analytics approach to create waste generation profiles for waste management and collection. Waste Manage. 9.

[15]. Alqahtani, F., Al-Makhadmeh, Z., Tolba, A., Said, W., 2020. Internet of things-based urban waste management system for smart cities using a Cuckoo Search Algorithm. Cluster Computing 23 (3), 1769–1780.

[16]. Idwan, S., Mahmood, I., Zubairi, J.A., et al.: Optimal management of solid waste in smart cities using internet of things. Wirel. Pers. Commun. 110, 485–501 (2020). https://doi.org/10.1007/ s11277-019-06738-8

[17]. Ruiz V., Sa´nchez A´ ., Ve´lez J.F., Raducanu B.: Automatic imagebased waste classification. In: Ferra´ndez Vicente J., A´ lvarezSa´nchez J., de la Paz Lo´pez F., Toledo Moreo J., Adeli H. (eds) From Bioinspired Systems and Biomedical Applications to Machine Learning. IWINAC 2019. Lecture Notes in Computer Science, vol. 11487. Springer, Cham (2019)

[18]. Misra, D., Das, G., Chakrabortty, T., et al.: An IoT-based waste management system monitored by cloud. J. Mater. Cycles Waste Manag. 20, 1574–1582 (2018).

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